

EARLY DETECTION OF PERIPHERAL IV INFILTRATION/EXTRAVASATION

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PROBLEM AND ORIGINAL GOAL

Problem:

- Late detection causes tissue injury
- Visual inspection may be delayed
- Need earlier and more objective monitoring

Proposed solution:

- Smartphone + thermal camera
- Monitor short clips around the IV site
- Detect abnormal thermal change early
- Work offline on-device

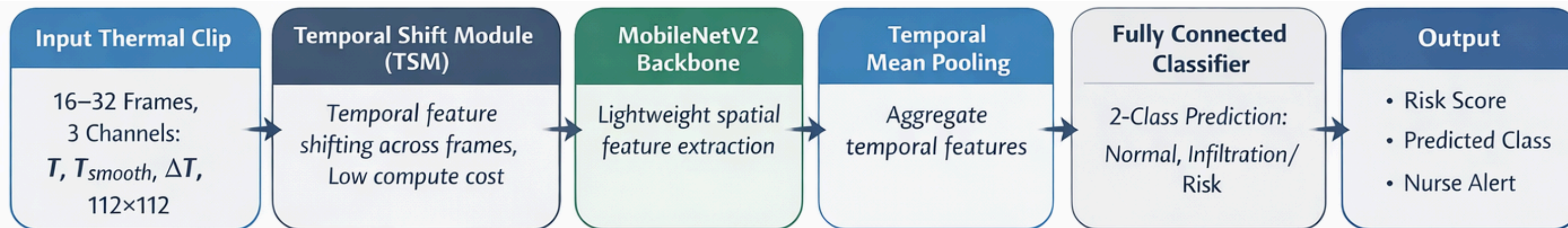
Goal: detect risk before visible swelling becomes obvious

PROGRESS SO FAR

1. Built the model using TSM-MobileNetV2
2. Built the data pipeline
3. Created synthetic data
4. Exported the model to ONNX and tested with ONNX Runtime
5. Started building the Android app with Kotlin

MODEL ARCHITECTURE

- Lightweight architecture suitable for mobile deployment
- Captures temporal information from short thermal clips
- Better fit for smartphone inference than heavier video models



Input shape: $3 \times T \times 112 \times 112$ → Lightweight temporal modeling → Mobile deployment target.

DATA PIPELINE AND SYNTHETIC DATA

Clip/video input → Preprocessing → Channel construction → Resize / normalize → Model input tensor

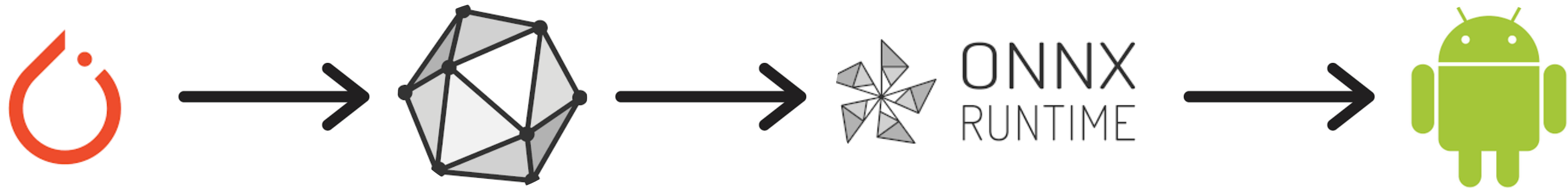
Key tasks:

- Extract short clips
- Prepare thermal inputs
- Organize frames for the model

Synthetic Data:

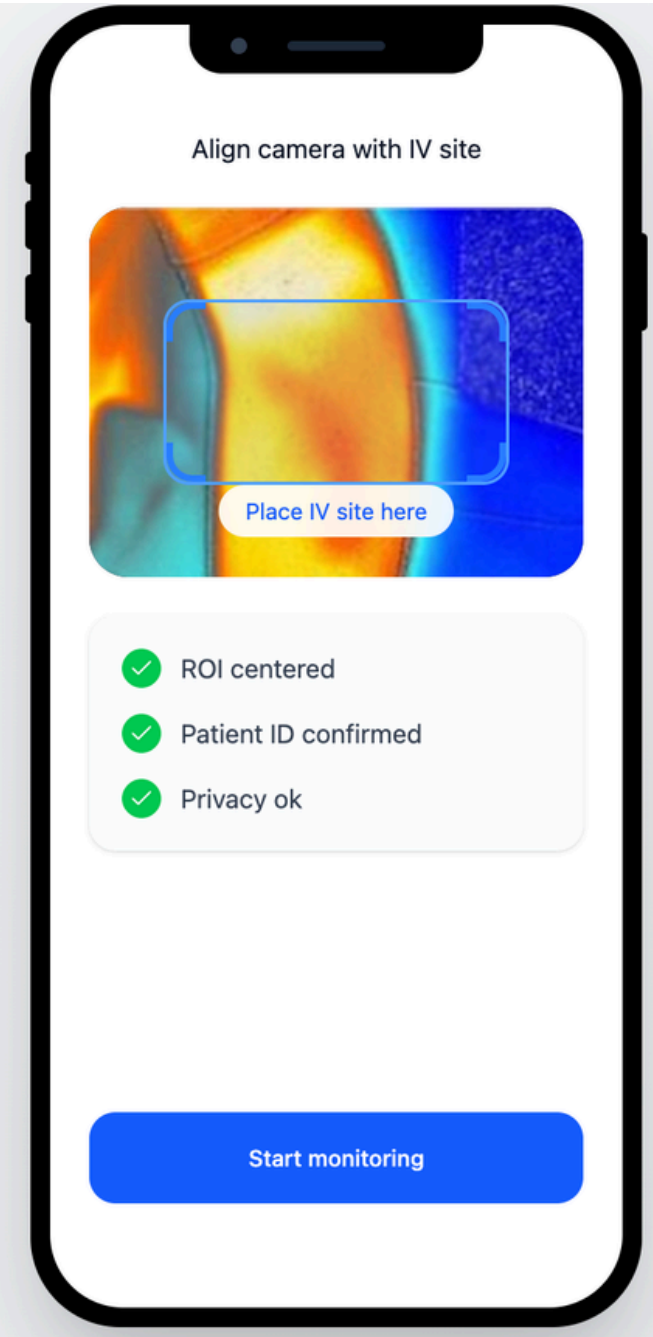
- Generated synthetic examples for development
- Helpful before full real clinical dataset is available
- Used to test training and inference workflow

DEPLOYMENT

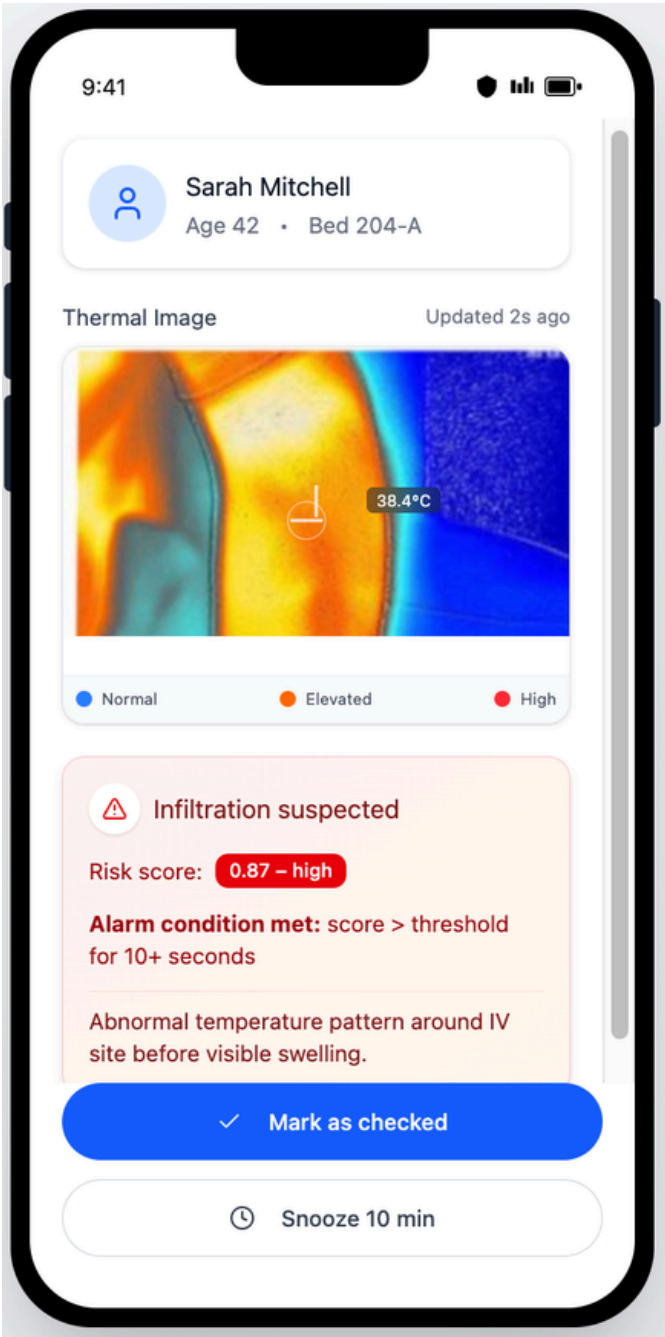


- Portable deployment format
- Supports offline inference
- Good step toward mobile integration

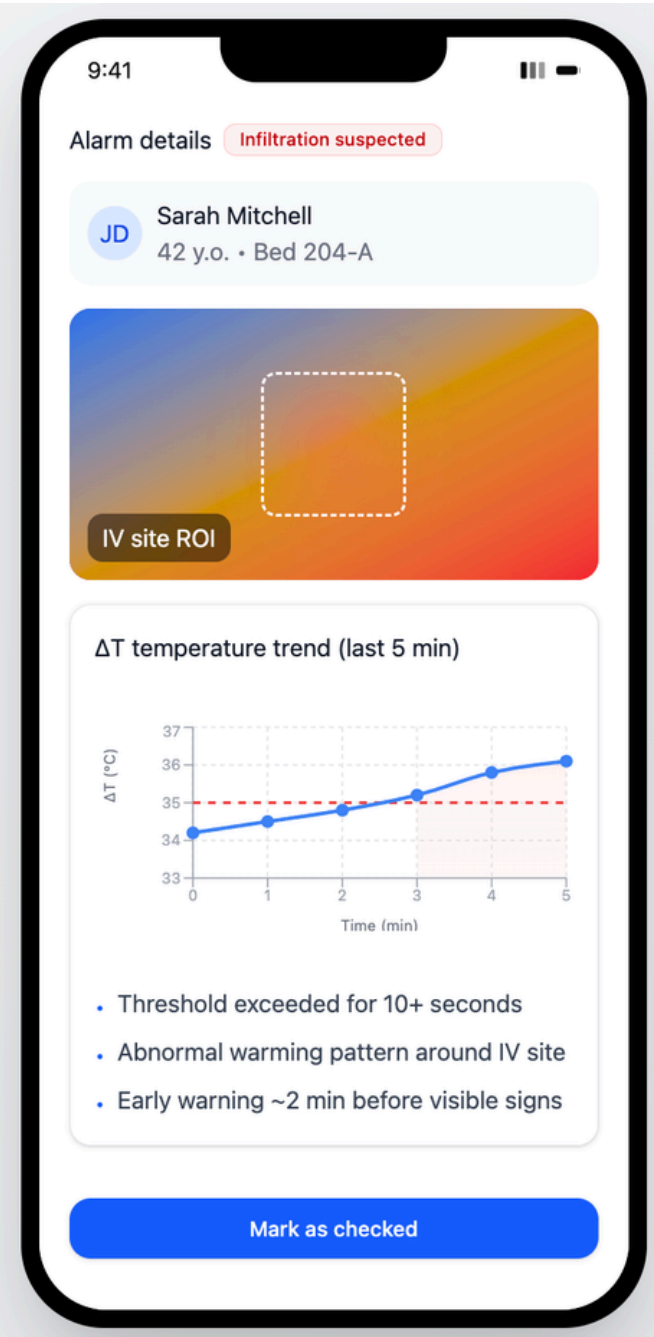
MOBILE APP PROGRESS



Setup



Monitoring



Result

THANK YOU

